CSE396

COMPUTER ENGINEERING

PROJECT

GROUP-3 PROJECT TUTORIAL

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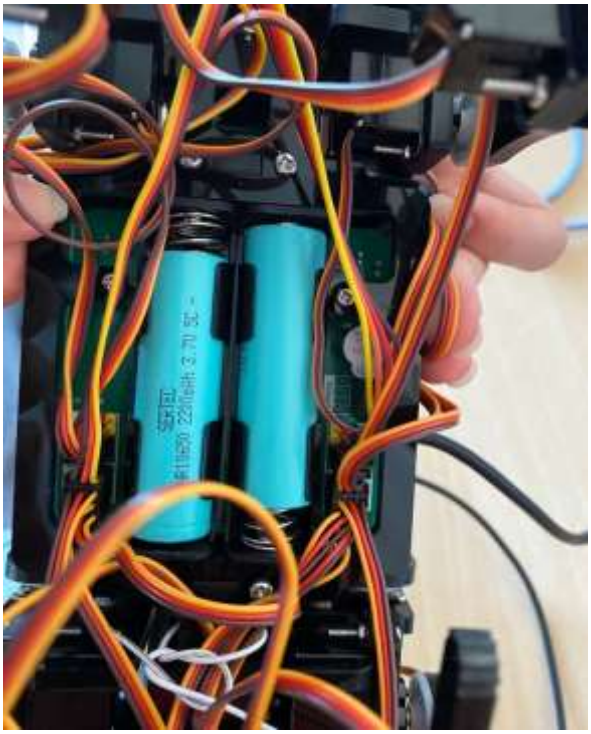
Priority Things To Do

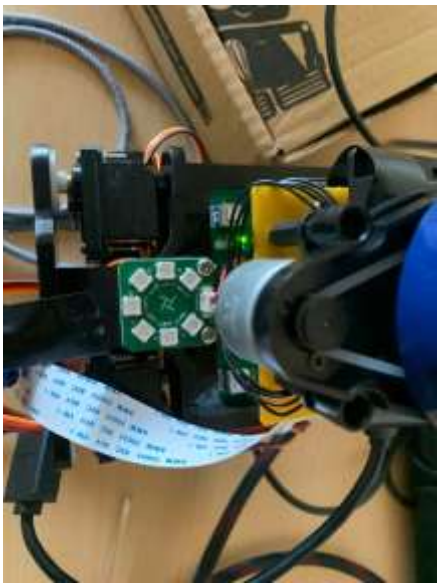
First, make sure that the robot's batteries are correctly inserted into the battery slot. Batteries should be installed as shown in the figure below:



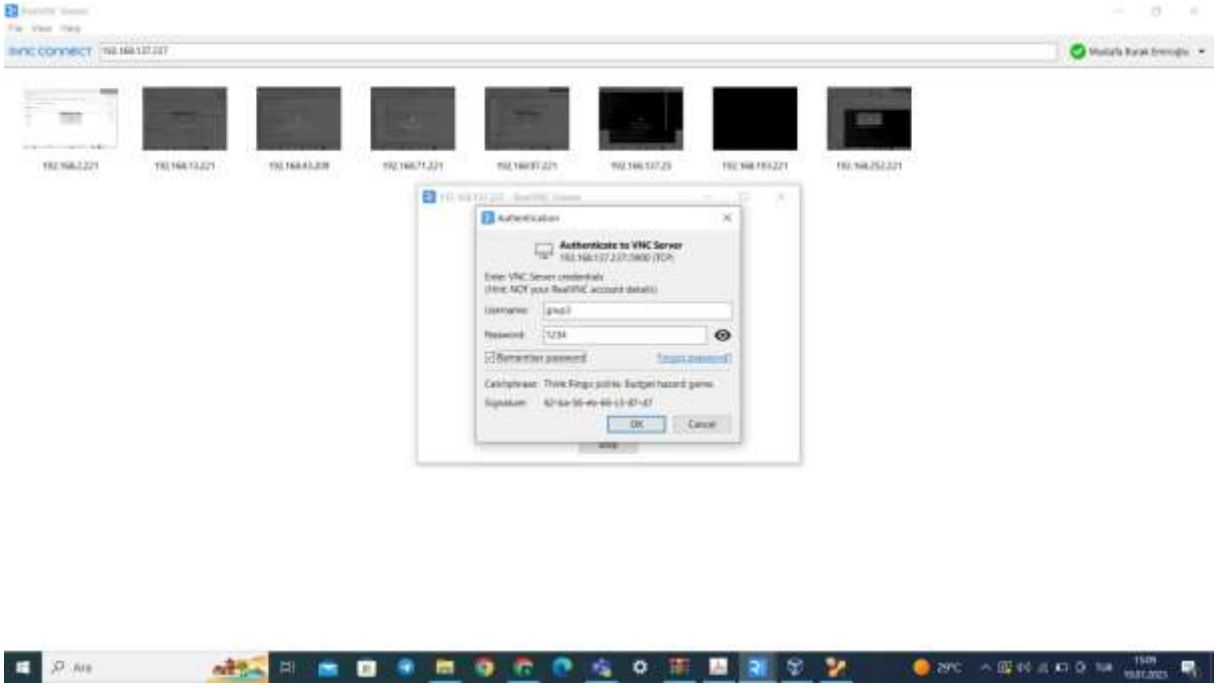
After the batteries are installed, the CTRL key on the robot should be pressed and then the LOAD key should be pressed.

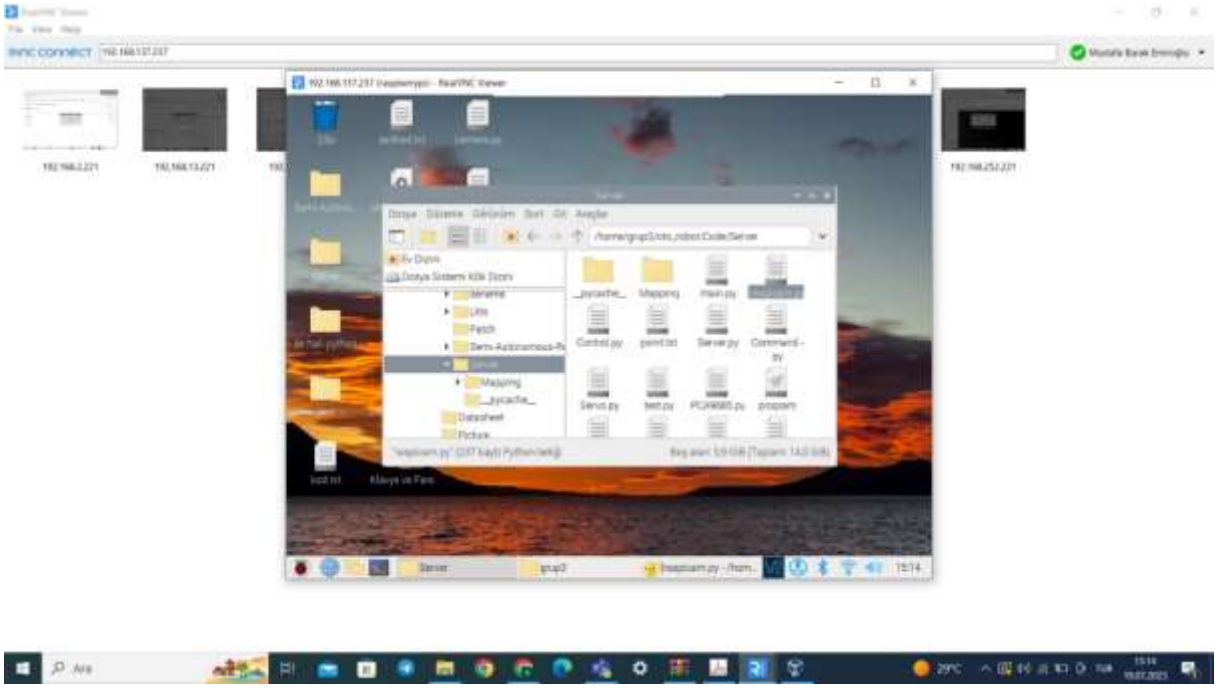
HDMI connection must be made from RaspberryPi HDMI input.Then connect lidar sensor from type-C enrty to raspberry pi.

After successful connection, internet connection is required.



With the VNC Viewer program, a connection is established from the IP address of the robot that has an internet connection. After connecting with VNC Viewer, you can disconnect the HDMI cable.



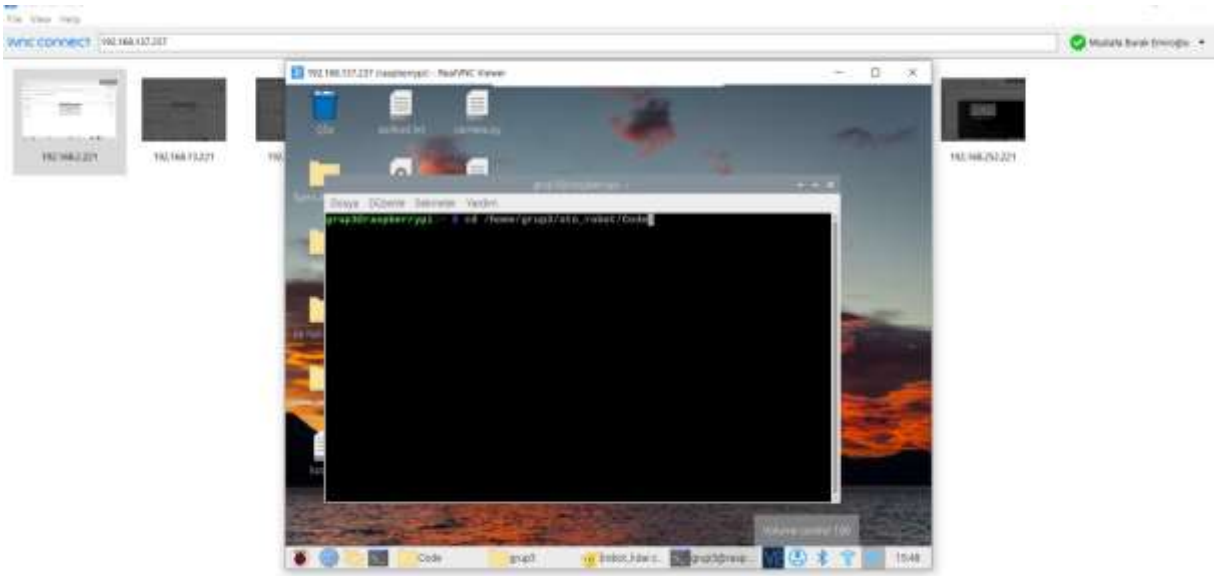


Important Note: With the same internet that RaspberryPi is connected to, both the server (the computer you connect with VNC Viewer) and the client side must be connected to the same internet.

SERVER SIDE

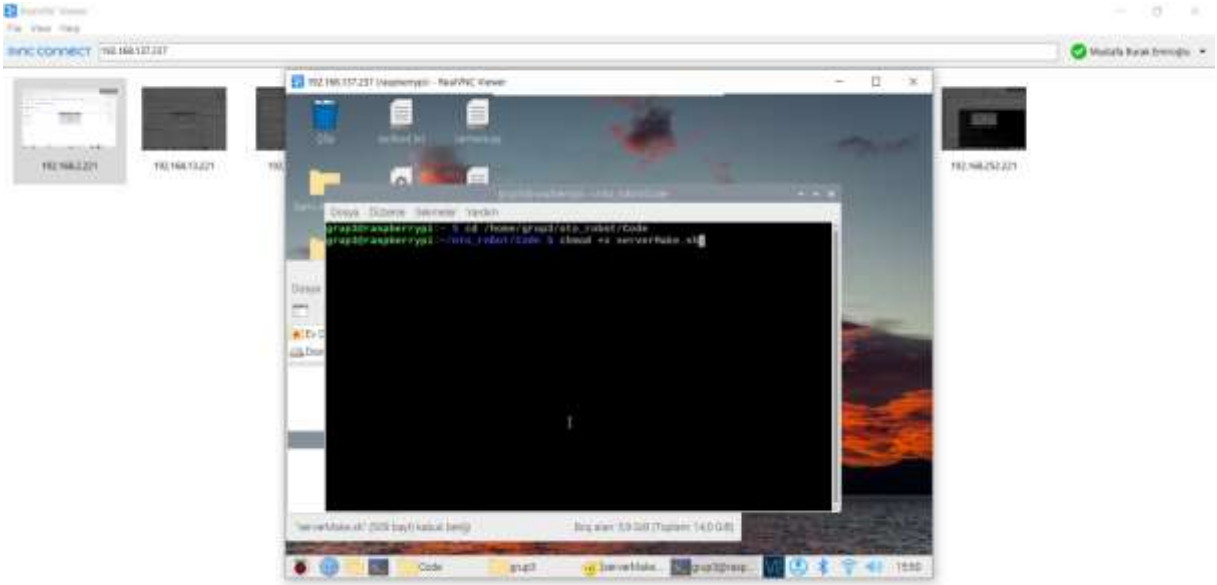
To run the server side of the program, the code on the server side must be run via VNC Viewer. Then you should enter the following path:

cd /home/grup3/oto_robot/Code



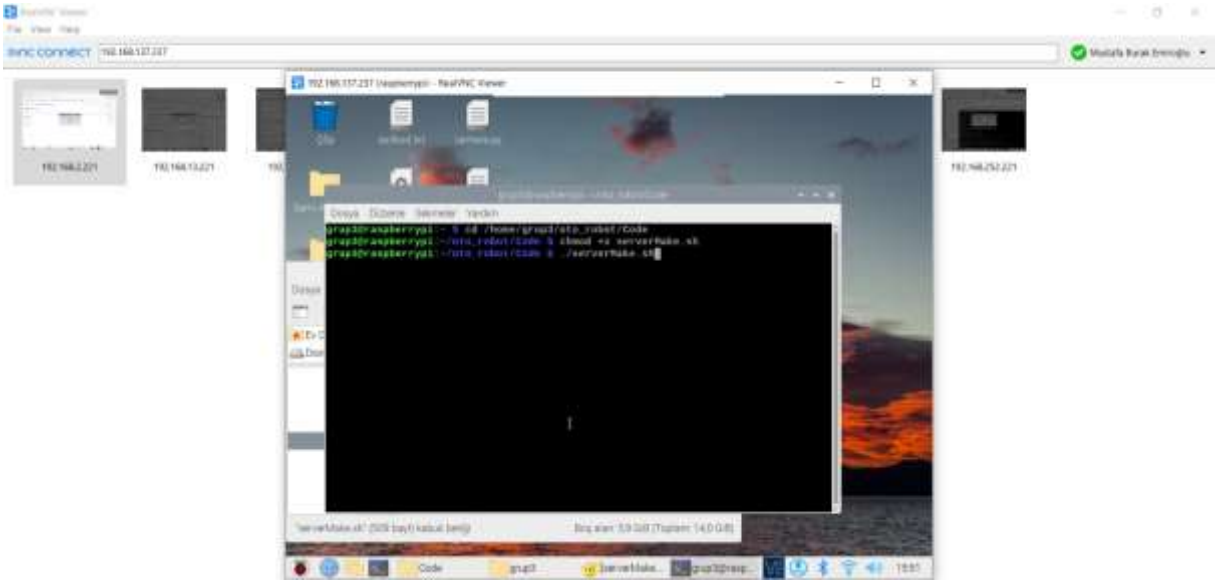
Run the following command to compile the bash script found here.

chmod +x serverMake.sh

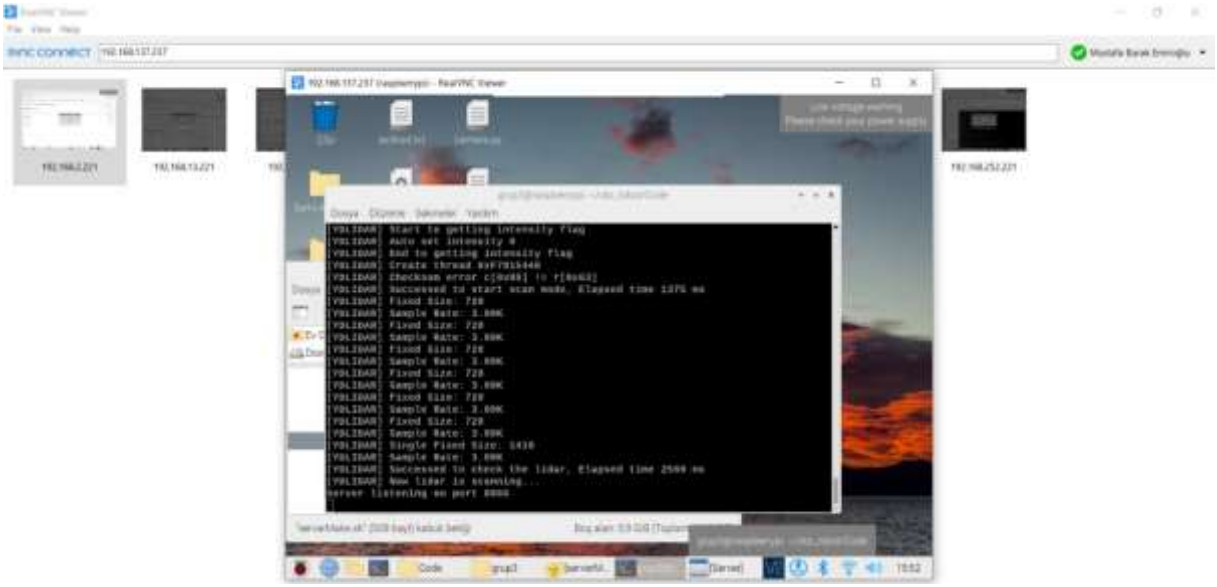


You can run the server program using the following command.

./serverMake.sh



You can see that the server program is running successfully.



Note: Please remember port number here. Because it should be same port in the client side.

Important: In case port number <u>is not</u> displayed on the screen, change port number in the following directory; /Server/Mapping/robot/robot_lidar.cpp

```
int main(int argc, char *argv[])
{

  int remote_port = 8066;

```

Change the remote_port variable and then execute **./serverMake.sh** again to run server.

Remember this port number should be updated on client directory in the **ui_client.py** file.

CLIENT SIDE

In order to run the client-side codes, a computer with the Ubuntu operating system is required. On this computer, you need to download our client codes given the github link below.

https://github.com/elif-ay/Semi-Autonomous-Robot-Device/tree/finalVersion

In addition, some libraries and dependencies required for this project must be downloaded to your system.

Open a terminal and run the following commands in order and make sure that each one is successfully installed.

- **sudo apt install python3-pip**
- **pip install yolov5**
- **pip install keras**
- **pip install tensorflow**
- **pip uninstall oepncv-python**
- **pip install opencv-python-headless**

Note: If you get an error about "qt" when you try to run "python3 Main.py", run this command in terminal: "pip3 install pyqt5".

Optional: In the demo, we used Nvidia's CUDA technology to make image processing and mapping faster when we run it on our own computer. If you want to use CUDA, it must be on your computer. If not and you want to use, "https://developer.nvidia.com/cuda-downloads" Go to this link and choose the one suitable for your computer and install it.

After installing the library and dependencies, you need to run client codes.

Firstly, the server and client are connected to each other over the network. For this reason, Raspberry and remote laptop must be on the same WiFi network with each other. Then you need to find out which IP address Raspberry has because you need to replace this IP address with the IP addresses in the client.

Run this command for learning IP address of Rpi on Rpi:

"hostname -I"

Then change this IP address and port number in the sections shown below in the **Main.py** and **ui_client.py** files in the Client directory. Port number should be same port number with server side. It's written in server's terminal.

```
23        self.robot_y = robot_y
24        self.robot_angle = robot_angle
25        self.ipAddress = "192.168.137.237"
26        self.remote_port = 8066
27        print("girdi3")
28
```

Figure-1:ui_client.py

```
91
92        self.stream_url = "http://192.168.137.237:8090"
93        self.cap = cv2.VideoCapture(self.stream_url)
94
```

Figure-2:Main.py

Open the terminal and go to the folder with the downloaded codes with the cd command.

```
ademmurat@ademmurat-GL553VD: ~
ademmurat@ademmurat-GL553VD:~$ cd /home/ademmurat/Desktop/project/Semi-Autonomous-Robot-Device-finalVersion/Client/
```

To run the user interface program, you need to run the following command.

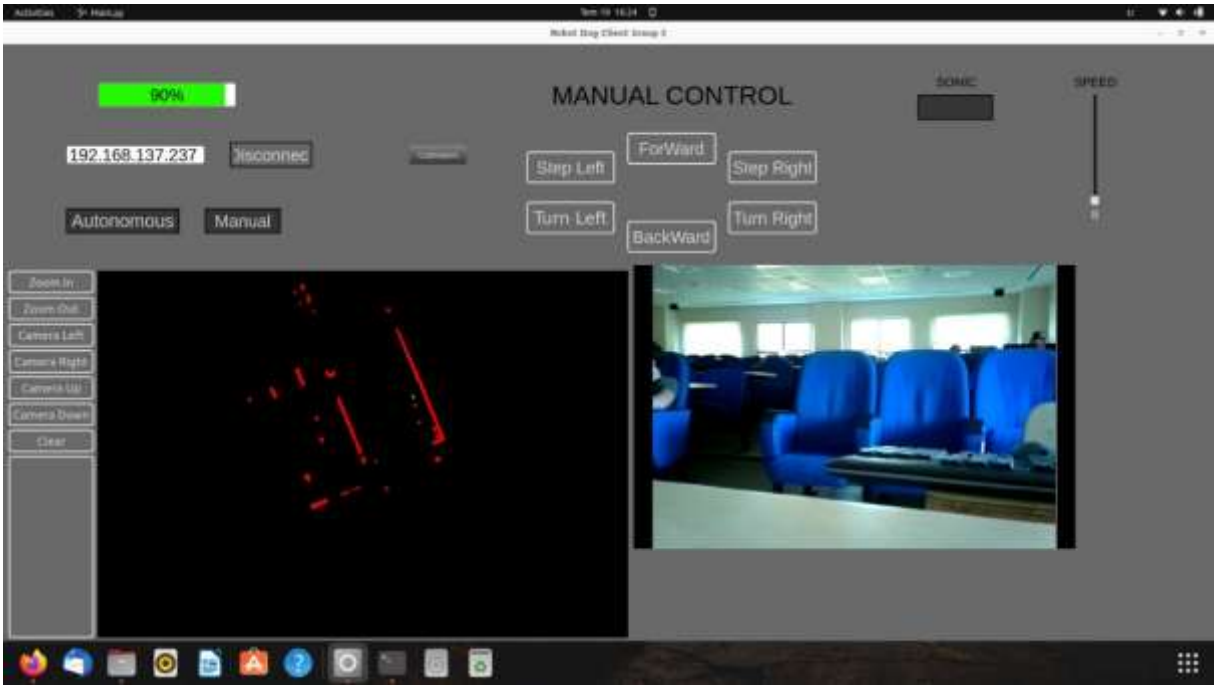
python3 Main.py

```
ademmurat@ademmurat-GL553VD:~$ cd /home/ademmurat/Desktop/project/Semi-Autonomous-Robot-Device-finalVersion/Client/
ademmurat@ademmurat-GL553VD:~/Desktop/project/Semi-Autonomous-Robot-Device-finalVersion/Client$ python3 Main.py
```

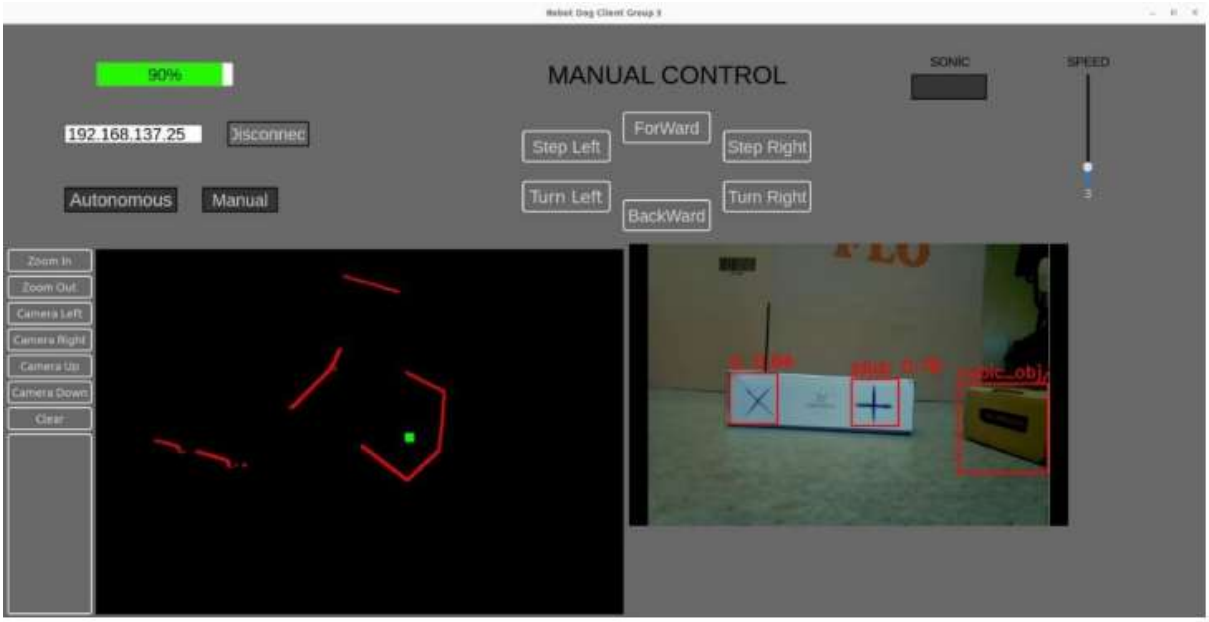
After running this command, you will see the following interface.

There are 6 buttons on the top-left side. The forward button allows the robot to step straight forward. Backward button makes it step backwards. Left and right buttons make the robot go left and right respectively. The step right and step left buttons also allow the robot to step right and left, respectively.

The Autonomus button on the right initiates the robot's autonomous movement. When you press this button, the robot starts to move straight. When there is an obstacle in front of him, he moves in the direction that is empty from his right and left. In this way, he can walk on his own without outside interference and complete the course. The Connect button on the upper left allows the robot to connect to the server. Make sure the IP address is the server's IP address and click the Connect button. In this way, a successful connection will be established.



There are two frames at the bottom. It shows the image of the map we extracted with the mapping algorithm on the left. On the right is the image transferred as a result of processing the camera image taken from Raspberry with the object detection algorithm. In this way, the robot can recognize the objects in front of it.



To terminate the process; First clode the client window then terminal.

After closing client window, you can close server.